Overview of the City Parking Data Dashboard

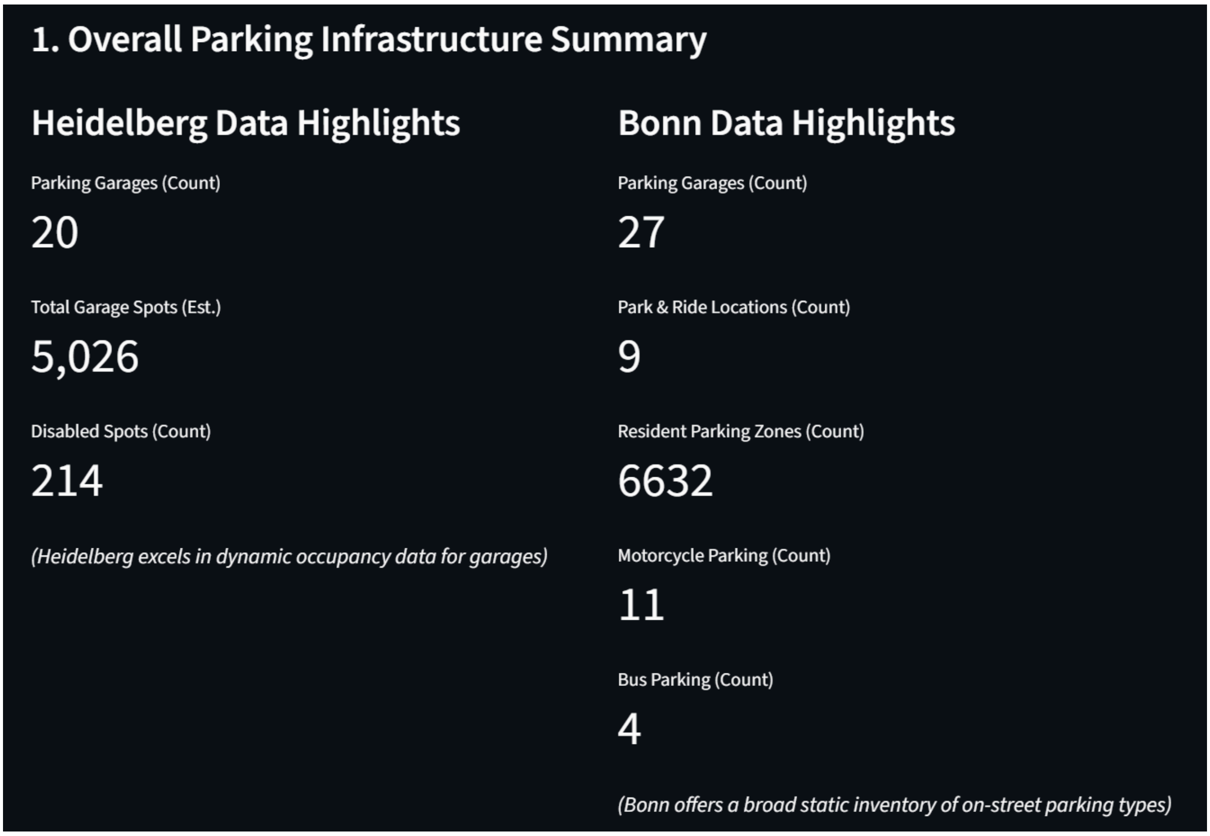
This dashboard serves as a vital tool for comparing the parking infrastructure and open data strategies of Heidelberg and Bonn. Its core objective is to highlight Heidelberg's strengths, identify areas where its open data portal can be enhanced, and propose actionable recommendations, drawing lessons from Bonn's established practices.

Dashboard Sections: A Comprehensive Guide

The dashboard is intuitively organized into distinct sections, each accessible through a convenient sidebar, providing a complete picture of the urban parking data landscape:

1. Overall Summary

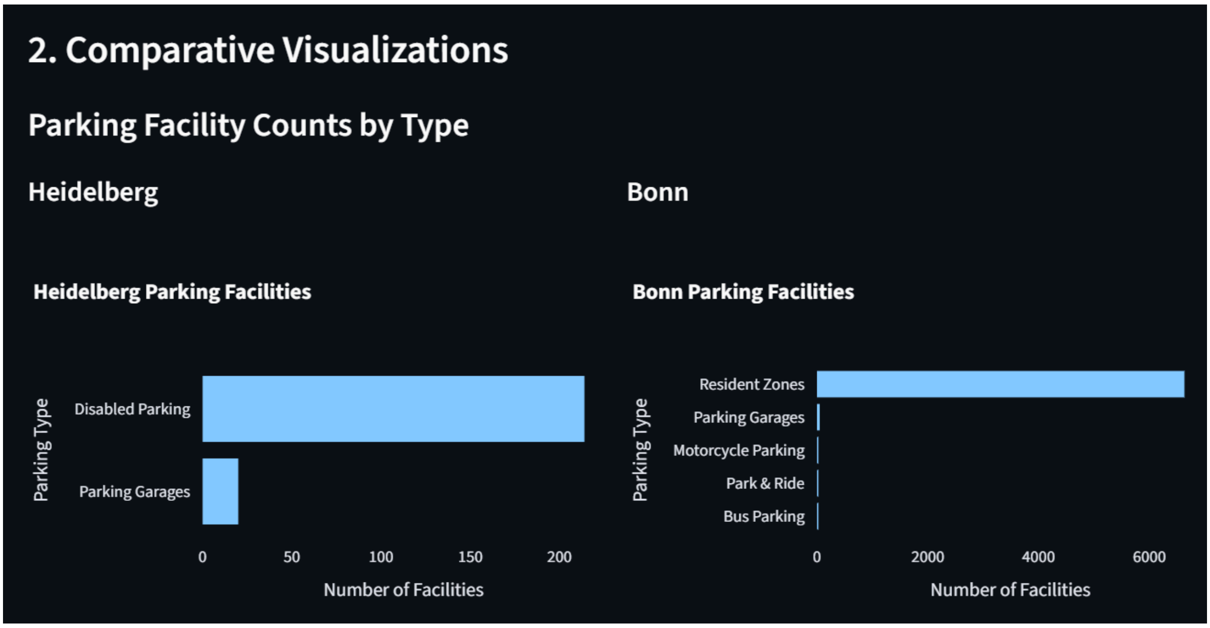
- **Purpose:** To provide a quick, high-level overview of the parking facilities in both Heidelberg and Bonn, presenting essential numbers to understand their scale at a glance.



- **Content:**
  - **Heidelberg Data Highlights:** Shows the total number of parking garages, an estimated total of available parking spots within those garages, and the count of dedicated disabled parking spaces. This section highlights Heidelberg's current strength in providing up-to-date information on garage occupancy.
  - **Bonn Data Highlights:** Presents the number of parking garages, Park & Ride locations, designated resident parking zones, motorcycle parking areas,

and bus parking areas. This illustrates Bonn's extensive and varied inventory of static on-street parking types.

- **Comparative Visualizations:**
  - **Purpose:** To visually compare the quantity of different types of parking facilities between the two cities.



  - **Presentation:** A clear bar chart displays the counts for key parking categories in both Heidelberg and Bonn. This visual comparison makes it easy to see the differences in parking infrastructure and diversity between the two cities.
- **Key Insights:** This section offers a concise summary of each city's parking profile, setting the stage for deeper comparisons.

2. Data Assets Overview

- **Purpose:** To offer a detailed inventory of the primary open parking datasets available from each city. This helps understand what raw information is publicly accessible and its general structure.

2. Data Assets: What Each City Provides

Below is an overview of the primary parking datasets available for each city, including their size and content summary.

Heidelberg Data Assets

Dataset Name	Rows	Columns	Description
parking_garage	20	42	Static information about pa
disabled_parking	214	12	Locations of dedicated disa
historical_p001	245521	21	Historical occupancy data (
current_p00	67658	12	Current/recent occupancy

(Heidelberg's strength: granular dynamic occupancy data for garages)

Bonn Data Assets

Dataset Name	Rows	Columns	Description
resident_parking_1	295	6	Geographical areas
resident_parking_2	6337	9	Geographical areas
park_and_ride	9	2	Locations of Park &
parking_garages	27	12	Static locations of p
general_parking	123	6	Comprehensive loc
bus_parking	4	7	Specific locations fo
motorcycle_parking	11	4	Specific locations fo
parking_bonn_koeln_osm	7805	151	OSM-derived parkin

(Bonn's strength: broad static inventory of on-street parking types)

- **Content:** Presents a table for each city, listing each available dataset along with:
  - **Dataset Name:** The specific name identifying the data collection.
  - **Number of Entries:** How many individual records or items are included in the dataset.
  - **Number of Information Fields:** How many different pieces of information are provided for each record.
  - **Description:** A straightforward explanation of what kind of parking data is contained within that dataset.
- **Key Insights:** This section helps stakeholders understand the current scope of open parking data in each city. It emphasizes Heidelberg's focus on data from parking garages and Bonn's broader collection of on-street parking details.

3. Dataset Attributes

- **Purpose:** To compare the specific types of information (attributes or data fields) that are recorded within each city's datasets. This comparison is vital for identifying opportunities to enrich Heidelberg's data by looking at what Bonn provides.
- **Content:** Displays a table for each city, showing all the unique attributes available across its datasets.
- **Key Insights:** By examining the listed attributes, stakeholders can quickly see which detailed pieces of information are present or missing in either city's data. This guides discussions on potential data enhancements, such as adding more specific parking restrictions or detailed capacity figures.

3. Dataset Attributes: What Information is Available

This section compares the attributes (columns/fields) present in each city's datasets. Observing Bonn's comprehensive attributes can help Heidelberg identify areas to enrich its data.

Heidelberg Dataset Attributes

Dataset	Attributes
parking_garage	_id, id, name, parkingSiteId, description, facilitie
disabled_parking	id, Notiz, BEZEICHNUN, BETREIBER, TYP, XTRID, l
historical_p001	_id, recvTime, fiwareServicePath, entityId, entity
current_p00	_id, recvTime, fiwareServicePath, entityId, entity

Bonn Dataset Attributes

Dataset	Attributes
resident_parking_1	bewohnerpark_id, parkgebiet_name, pa
resident_parking_2	parkgebiet, parkgebiet_name, komplett
park_and_ride	type, features
parking_garages	poi_id, name, plz, ort, strasse, hausnr, t
general_parking	parkplatz_id, bezeichnung, art, inhalt, l
bus_parking	parkplatz_id, bezeichnung, the_geom, e
motorcycle_parking	parkplatz_id, bezeichnung, longitude, la
parking_bonn_koeln_osm	@id, amenity, type, wheelchair, longitu

Comparing attributes highlights where Heidelberg might expand its data collection (e.g., more granular on-street restrictions, detailed facility info).

Heidelberg Dataset Attributes

Dataset	Attributes

parking_garage	_id, id, name, parkingSiteId, description, facilities, openingHours, closingHours, status, latitude, longitude, streetAddress, postalCode, addressLocality, addressCountry, areaServed, streetAddressDriveway, streetAddressExit, images, acceptedPaymentMethod, priceModelValidFor24h, numberOfPayingStations, dayStart, dayEnd, contactPointTelephone, contactPointEMail, provider, highestFloor, lowestFloor, specialLocation, maximumAllowedHeight, maximumAllowedWidth, totalSpotNumber, handicappedParkingSpots, womenParkingSpots, familyParkingSpots, maximumParkingDuration, forbiddenVehicleType, goodwillPeriodInMinutes, part, charging_stations, prices
disabled_parking	id, Notiz, BEZEICHNUN, BETREIBER, TYP, XTRID, URN, BESCHREIBU, BESCHRIFTU, PARKPLATZ_, longitude, latitude

historical_p001	_id, recvTime, fiwareServicePath, entityId, entityType, areaServed, maximumAllowedWidth, latitude, description, alternateName, observationDateTime, totalAvailableSpotNumber, occupiedSpotNumber, availableSpotNumber, name, maximumAllowedHeight, totalSpotNumber, aggregateRating, outOfServiceSlotNumber, longitude, status
current_p00	_id, recvTime, fiwareServicePath, entityId, entityType, name, observationDateTime, occupiedSpotNumber, parkingSiteId, status, totalSpotNumber, trend

Bonn Dataset Attributes

Dataset	Attributes
resident_parking_1	bewohnerpark_id, parkgebiet_name, parkgebiet_buchstabe, bereich, longitude, latitude
resident_parking_2	parkgebiet, parkgebiet_name, komplett_str, objekt, hausnr, strasse, langname, longitude, latitude

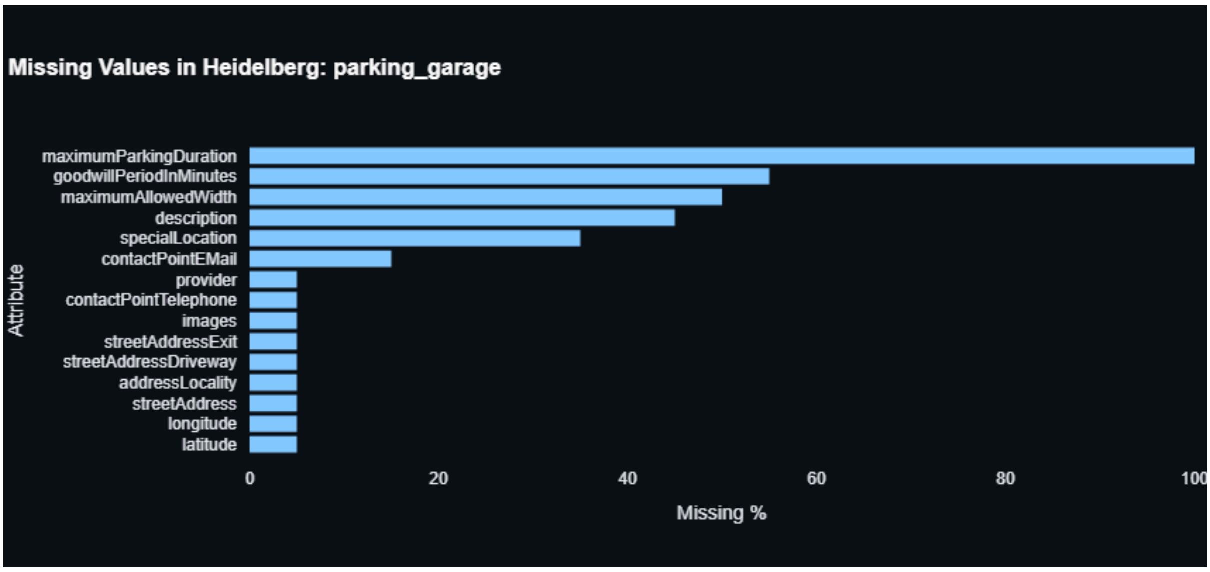
park_and_ride	type, features
parking_garages	poi_id, name, plz, ort, strasse, hausnr, tel, email, internet, parkhaus_id, longitude, latitude
general_parking	parkplatz_id, bezeichnung, art, inhalt, longitude, latitude
bus_parking	parkplatz_id, bezeichnung, the_geom, eigentum, art, longitude, latitude
motorcycle_parking	parkplatz_id, bezeichnung, longitude, latitude

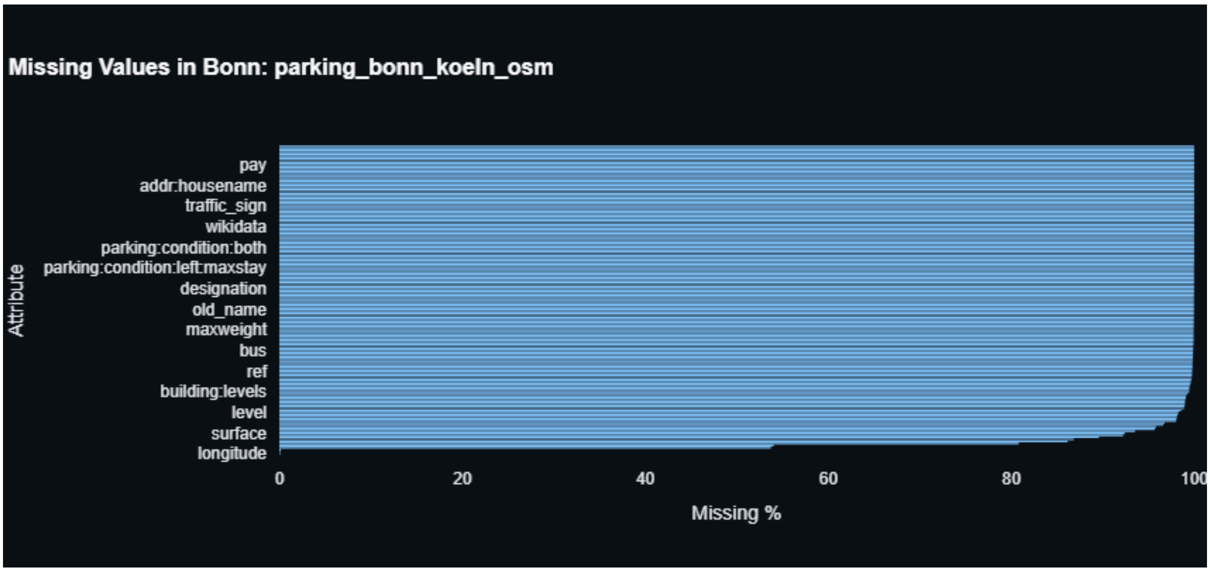
parking_bonn_koeln_osm	@id, amenity, type, wheelchair, longitude, latitude, access, building, capacity, fee, lit, name, opening_hours, operator, parking, source, layer, level, surface, capacity:disabled, capacity:women, height, maxheight, roof:material, building:levels, roof:shape, supervised, addr:city, addr:housenumber, addr:postcode, addr:street, addr:suburb, park_ride, toilets:wheelchair, maxstay, power_supply, tourism, ele, ele:DHHN92, created_by, description, barrier, smoothness, addr:country, max_time, ref, covered, building:use, coach, official_name:DB, railway:ref:DB, tourist_bus, vehicle, wheelchair:description, parking:condition:both, parking:condition:both:time_interval, parking:lane:both, levels, website, name:de, name:en, designation, motorcar, note, free_parking_time, foot, parking:condition:area, goods, hgv, maxweight, service, motorcycle, maxheight:physical, fixme, man_made, surveillance, emergency, psv, phone, wikidata, leisure, landuse, building:part, capacity:parent, free, caravans, smoking, FIXME, building:colour, building:material, roof:colour, old_name, area, alt_name, min_height, fee:conditional, bus, parking:condition, fee:times, maxstay:for_free, fenced, motor_vehicle, start_date, construction, traffic_sign, bridge, tunnel, building:ref, shop, highway, postal_code, stay, building:levels:underground, park_and_ride, building:min_level, parking:payment, parking:payment:time_interval,

capacity:bus, description:en,
note:de, disused, email, fax,
addr:housename,
parking:condition:left,
parking:condition:left:maxstay,
parking:condition:left:time_interval,
charge, note:fee, description:de,
source:outline, is_in:city,
is_in:postal_code, disused:name,
park, maxstay:conditional,
@relations, bicycle, horse, oneway,
maxspeed, pay,
payment:american_express,
payment:coins, payment:girocard,
payment:mastercard,
payment:notes, access:description,
access:conditional, noexit,
disabled_spaces

4. Data Quality Dashboard

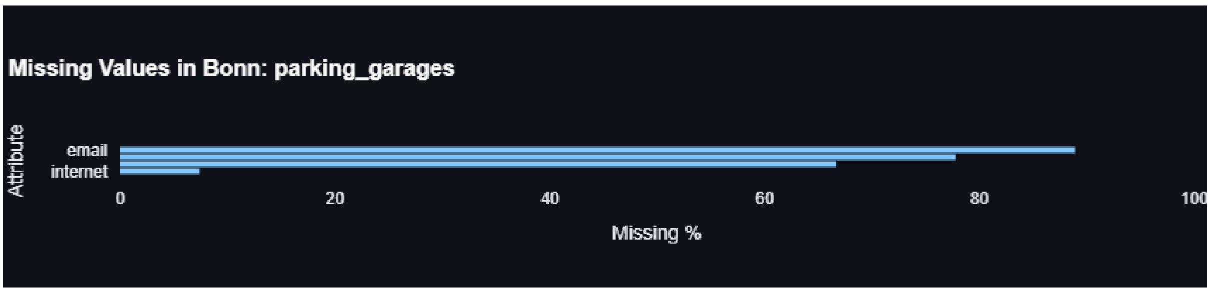
- **Purpose:** To provide a clear and straightforward assessment of how complete each dataset is, highlighting where information might be missing. Understanding data completeness is crucial for ensuring the reliability and usefulness of any analysis or application built upon it.
- **Content:** This section presents horizontal bar charts for each individual parking dataset from both Heidelberg and Bonn.
  - Each bar visually represents a specific piece of information (an attribute) within that dataset.
  - The length of the bar shows the percentage of records where that particular information is missing, making it easy to identify significant data gaps.


Missing Values in Heidelberg: parking_garage
Attribute
maximumParkingDuration
goodwillPeriodInMinutes
maximumAllowedWidth
description
specialLocation
contactPointEMail
provider
contactPointTelephone
images
streetAddressExit
streetAddressDriveway
addressLocality
streetAddress
longitude
latitude
0
20
40
60
80
100
Missing %

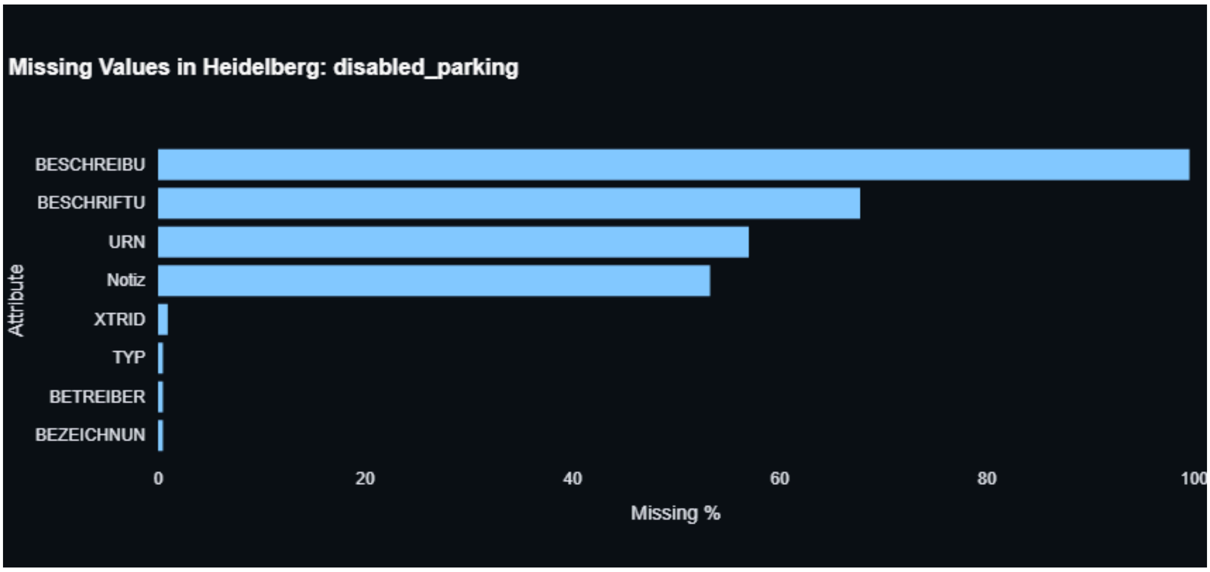

Missing Values in Bonn: parking_bonn_koeln_osm
Attribute
pay
addr:housename
traffic_sign
wikidata
parking:condition:both
parking:condition:left:maxstay
designation
old_name
maxweight
bus
ref
building:levels
level
surface
longitude
0
20
40
60
80
100
Missing %


Missing Values in Bonn: bus_parking
the_geom
0
20
40
60
80
100
Missing %


Missing Values in Bonn: general_parking
bezeichnung
0
20
40
60
80
100
Missing %


Missing Values in Bonn: parking_garages
Attribute
email
internet
0
20
40
60
80
100
Missing %


Missing Values in Bonn: resident_parking_2
komplett_str
0
20
40
60
80
100
Missing %



- **Key Insights:** These visuals quickly point out which datasets and their corresponding information fields have the most incomplete data. This informs decision-making regarding data collection priorities, validation processes, and overall data management strategies to improve quality.

5. Geographic Distribution

- **Purpose:** To allow users to visually explore the physical locations of parking facilities across both cities. This interactive map provides a spatial context, which is highly beneficial for urban planning, traffic management, and user navigation.
- **Content:** An interactive map that displays:
  - Markers on the map precisely indicating the locations of various parking facilities.
  - Distinct colors and symbols are used to differentiate between various types of parking (e.g., parking garages, disabled parking spots, Park & Ride areas).
  - Clicking on a location marker reveals a small pop-up with basic details about that specific parking facility.
- **Interaction:** The map is fully interactive, allowing users to move around, zoom in and out, and filter the displayed parking locations. Filters enable users to show only specific parking types (like "Disabled Parking") or focus on one city ("Heidelberg," "Bonn," or "Both Cities") at a time.
- **Key Insights:** This visual tool helps stakeholders understand the physical spread of parking resources. It also allows for observation of differences in how densely parking facilities are distributed and the variety of parking types made available on the map by each city's data.

6. Recommendations

- **Purpose:** To translate the findings from the comparative analysis into concrete, actionable suggestions for Heidelberg's open parking data initiatives.
- **Content:** A structured list of strategic recommendations. Each recommendation clearly outlines:
  - **The Issue:** A concise description of the challenge or area for improvement identified within Heidelberg's current data or strategy.
  - **The Recommendation:** A clear, practical suggestion for how to address the identified issue.
  - **The Benefit:** The anticipated positive outcome or value that will be gained by implementing the recommendation, illustrating its importance to the city.

6. Recommendations for Heidelberg's Open Data Portal

Based on the comparative analysis with Bonn's data, here are key recommendations for Heidelberg to enhance its open parking data:

1. Issue: Heidelberg's current data lacks granular detail for various on-street parking types (e.g., resident zones, motorcycle, bus parking).

Recommendation: Introduce new datasets and categories for on-street parking types, similar to Bonn's comprehensive approach.

Benefit: Provides a more complete picture of the city's overall parking infrastructure, aiding urban planning and diverse user navigation.

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2. Issue: Explicit capacity information for on-street parking is often missing in Heidelberg's datasets.

Recommendation: Include `capacity` attributes consistently across all new and existing on-street parking datasets.

Benefit: Enables quantitative analysis of on-street availability and improves resource allocation for better parking management.

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3. Issue: Dynamic occupancy data is currently limited to parking garages.

Recommendation: Investigate and implement methods (e.g., sensors) to collect and publish real-time/historical occupancy data for key on-street parking areas.

Benefit: Extends Heidelberg's existing strength in dynamic data to the on-street environment, significantly enhancing utility for drivers seeking real-time information.

4. Issue: Potential data gaps in critical fields (e.g., location coordinates, identifiers) across datasets.

Recommendation: Implement regular data quality checks and robust validation processes to ensure the completeness and reliability of all open data.

Benefit: Guarantees higher reliability and usability of all open data for applications, analyses, and informed decision-making.

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5. Issue: Maximize the impact and value derived from open data initiatives.

Recommendation: Actively highlight and promote the diverse types of parking data available (both existing and newly added) to encourage third-party app development, academic research, and public engagement.

Benefit: Increases the overall impact and value derived from Heidelberg's open data portal, fostering innovation and better citizen services.

- **Key Insights:** This section directly leverages the dashboard's analysis to provide a roadmap for enhancing Heidelberg's open data portal. It covers vital areas such as expanding on-street parking data, including capacity details, exploring dynamic occupancy data for wider areas, strengthening data quality assurance, and actively promoting data use for innovation and improved citizen services.

This dashboard and its detailed documentation aim to equip stakeholders with the necessary insights to make informed decisions and strategically plan for the future of urban parking data in Heidelberg.